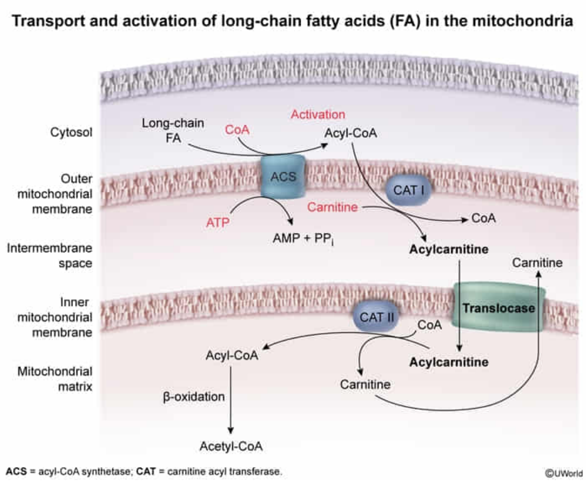
In order to transport long-chain fatty acids from the cytosol to the mitochondria, three of the following molecules are required. Which one is NOT necessary?

- A. Coenzyme A [12%]
- B. ATP [18%]
- C. Carnitine [14%]
- D. Oxaloacetate [55%]

Correct
56% answered correctly

Explanation:



Entry into the mitochondrial matrix is **tightly regulated** by the **inner mitochondrial membrane**. Crossing the membrane generally requires the aid of transport proteins, which only recognize specific molecules. Some short-chain fatty acids can cross the inner mitochondrial membrane directly, but long-chain fatty acids can only cross the outer mitochondrial membrane.

Additionally, long-chain fatty acids are not recognized by any transport proteins; they must be modified or **"activated" to enter the matrix**. Fatty acid activation and transport includes the following steps:

- In the cytosol, the enzyme acyl-CoA synthetase catalyzes the reaction of fatty acids with **coenzyme A** to form acyl-CoA molecules. This reaction is thermodynamically unfavorable and requires **ATP hydrolysis** to proceed **(Choices A and B)**.

- Acyl-CoA molecules then migrate to the intermembrane space, where they can react with **carnitine** to form acylcarnitine. The transport protein acylcarnitine translocase, located on the inner mitochondrial membrane, recognizes acylcarnitine and carries it into the mitochondrial matrix **(Choice C)**.

- Acylcarnitine is then converted back to acyl-CoA and carnitine. Acyl-CoA is subsequently digested by β-oxidation, and carnitine is transported out of the matrix where it can pick up and carry a new fatty acid into the mitochondrial matrix.

Although **oxaloacetate** is an intermediate in the citric acid cycle, gluconeogenesis, and transport of acetyl-CoA *from* the mitochondria into the cytosol for fatty acid synthesis, it is **not involved in transport** of fatty acids from the cytosol *into* the mitochondria.

Educational objective:
Entry into the mitochondrial matrix is tightly regulated by transport proteins in the inner mitochondrial membrane. Fatty acids must be activated with coenzyme A followed by carnitine to enter the mitochondrial matrix. Activation requires ATP hydrolysis.

Time spent:0 sec
Subject:
Biochemistry

QID:402044
System:
Metabolic Reactions